

1.6 Ethical, legal, cultural and environmental impacts of digital technology

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 Teacher guide
 Unit 1.6
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Where this unit sits, what it covers, and how far

Where this unit sits

Qualification	GCSE Computer Science (9-1), specification code J277, from 2020
Component	Component 01: Computer systems — 80 marks, 1 hour 30 minutes
What OCR says the component covers	the central processing unit (CPU), computer memory and storage, data representation, wired and wireless networks, network topologies, system security and system software
This unit	topic 1.6 Ethical, legal, cultural and environmental impacts of digital technology
Checked	3 October 2026, at https://www.ocr.org.uk/qualifications/gcse/computer-science-j277-from-2020/specification-at-a-glance/

How far this has been checked. The topic numbers and titles below are the ones this course uses on its worksheets and slides. They were not re-checked against the published specification document when this map was built: the document could not be retrieved from this build environment. Check them against <https://www.ocr.org.uk/Images/558027-specification-gcse-computer-science-j277.pdf> before relying on this map for a scheme of work, and treat any finer subdivision as absent rather than assumed - no sub-point numbers are claimed anywhere in these documents, because none were verified.

Before it: 1.1 Systems architecture, 1.2 Memory and storage, 1.3 Computer networks, connections and protocols, 1.4 Network security, 1.5 Systems software

After it: 2.1 Algorithms, 2.2 Programming fundamentals, 2.3 Producing robust programs, 2.4 Boolean logic, 2.5 Programming languages and Integrated Development Environments, PY Python programming

The lesson sequence and the questions it answers

#	LESSON	THE BIG QUESTION	ACTIVITIES	MARKS
1	Impact desk e1-101	What should be considered before a new technology is introduced?	13	33
2	Privacy lab e1-102	What privacy issues does computing create, and where should the limits be?	13	25
3	Law court e1-103	What does computing legislation prohibit?	12	25
4	Culture street e1-104	How does computing affect how people live and work?	13	30
5	Planet works e1-105	What is the environmental impact of computing, across a device's whole life?	13	29
6	Stakeholder room e1-106	How does digital technology affect society locally, nationally and globally?	14	27

#	LESSON	THE BIG QUESTION	ACTIVITIES	MARKS
7	Licence lab e1-l07	What are the implications of choosing open source or proprietary software?	13	28
8	End of topic test e1-l08-test	assesses the whole unit	0	0
9	Name that Act e1-bonus	stretch work, no new content	1	0

Prior knowledge, vocabulary and skills

Assumed before the unit: the outcomes of 1.1, 1.2, 1.3, 1.4, 1.5

Vocabulary introduced: Ethical issue, Stakeholder, Personal data, Consent, Data Protection Act 2018, Computer Misuse Act 1990, Digital divide, Globalisation, E-waste, Carbon footprint, Trade-off, Open source, Proprietary

What a pupil does, not only reads: impact, lawgame, match, mcq, multi, order, sort

Coverage matrix

Every outcome this unit states, and whether it is taught, practised and assessed **within this unit**. Full means all three; partial means some; not covered means none of the three could be traced.

How this was worked out. The trace is derived by matching the content words of an outcome against the station text, the activities and the exam questions of the unit, not authored by hand. The words each match was made on are in build/coverage . json. It shows where to look; it does not certify that the alignment is right, and a full row has not been checked by a person either.

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
know the range of issues to consider when a new technology is introduced e1-l01-o1	e1-l01	yes	yes	yes	Full
understand at least one ethical issue of computer technology e1-l01-o2	e1-l01	yes	yes	yes	Full
be able to argue both sides and reach a conclusion e1-l01-o3	e1-l01	yes	yes	yes	Full
understand at least one issue related to privacy and computer technology e1-l02-o1	e1-l02	yes	yes	yes	Full
be able to weigh the benefits and risks of collecting personal data e1-l02-o2	e1-l02	yes	yes	yes	Full
know the principles of the Data Protection Act 2018 e1-l03-o1	e1-l03	yes	yes	yes	Full
know the principles of the Computer Misuse Act 1990 e1-l03-o2	e1-l03	yes	yes	yes	Full
know the principles of the Copyright, Designs and Patents Act 1988 e1-l03-o3	e1-l03	yes	yes	yes	Full
understand the impact of technology on daily life e1-l04-o1	e1-l04	yes	yes	yes	Full
understand the digital divide e1-l04-o2	e1-l04	yes	yes	yes	Full

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
understand globalisation e1-l04-o3	e1-l04	yes	yes	yes	Full
understand the environmental impact of manufacturing computers e1-l05-o1	e1-l05	yes	yes	yes	Full
understand the environmental impact of using computers e1-l05-o2	e1-l05	yes	yes	yes	Full
understand the environmental impact of disposing of computers e1-l05-o3	e1-l05	yes	yes	yes	Full
be able to identify key stakeholders e1-l06-o1	e1-l06	yes	yes	yes	Full
be able to consider a scenario from the perspective of stakeholders e1-l06-o2	e1-l06	yes	yes	yes	Full
understand impacts at local, national and global levels e1-l06-o3	e1-l06	yes	yes	yes	Full
understand the difference between open source and proprietary software e1-l07-o1	e1-l07	yes	yes	yes	Full
understand the implications of using open source and proprietary software e1-l07-o2	e1-l07	yes	yes	yes	Full

19 full · 0 partial · 0 not covered, of 19 outcomes.

Gaps, assumptions and overlap

None found by the trace.

What could not be traced

The wall text of e1-l01, e1-l02, e1-l03, e1-l04, e1-l05, e1-l06, e1-l07, e1-bonus is painted into the 360 image and held nowhere as text, so an outcome in those lessons lesson can only be traced through its activities and its exam questions. Read the rows above for them with that in mind.

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